

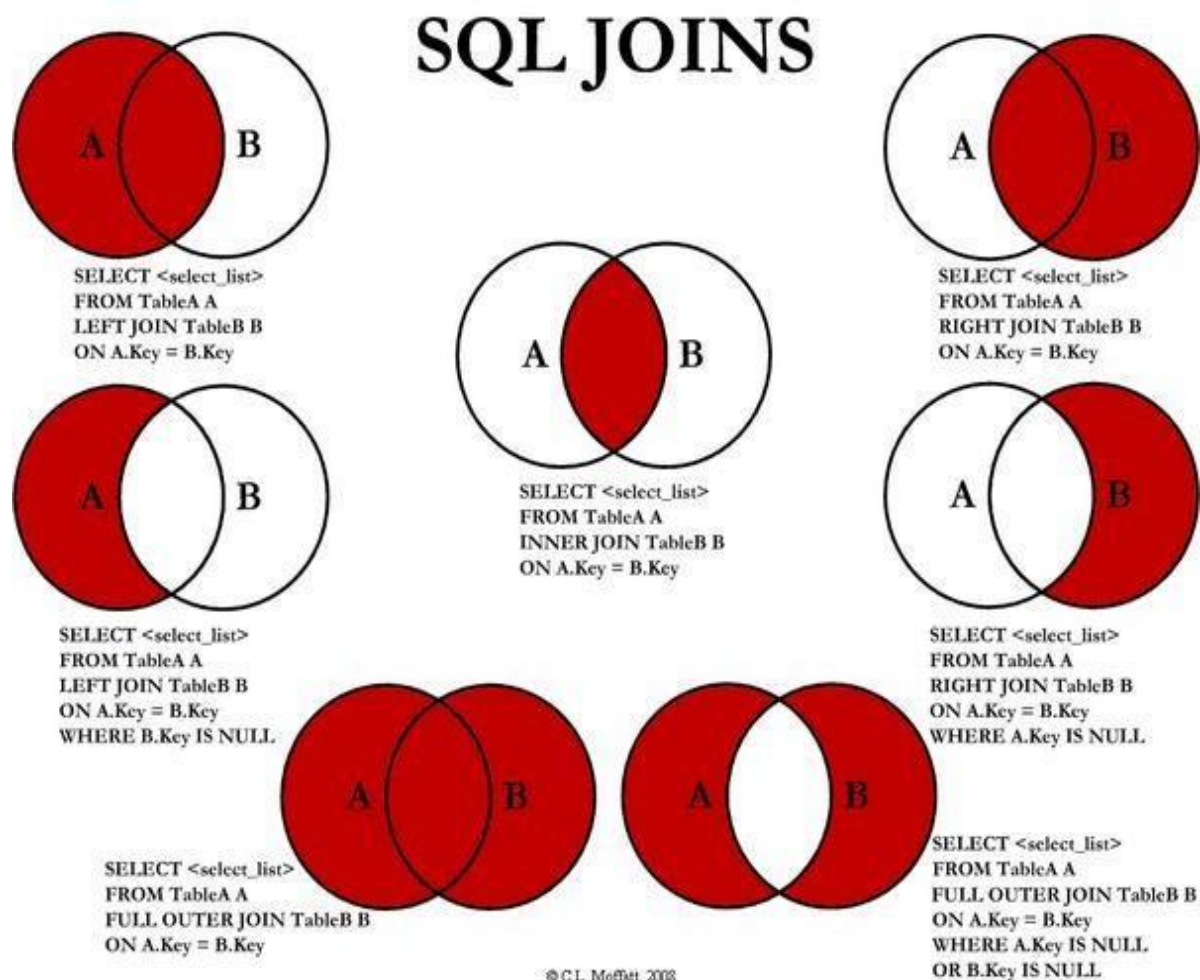
DBMS Lab Assignment – 5

SQL Joins

SQL JOINS are used to retrieve data from multiple tables. A SQL JOIN is performed whenever two or more tables are listed in a SQL statement.

There are 4 different types of SQL joins:

- SQL INNER JOIN (sometimes called simple join)
- SQL LEFT OUTER JOIN (sometimes called LEFT JOIN)
- SQL RIGHT OUTER JOIN (sometimes called RIGHT JOIN)
- SQL FULL OUTER JOIN (sometimes called FULL JOIN)



Assignment 5 Lab Questions

Using SQL joins, obtain the following information from SIU Library Database

1. Students in the IT department.

```
mysql> SELECT s.Stuid, s.Sname, s.email
-> FROM student s
-> JOIN department d ON s.deptid = d.Deptid
-> WHERE d.deptname = 'IT';
```

Stuid	Sname	email
805	Rohan Deshmukh	rohan.d@student.siu.edu

1 row in set (0.01 sec)

2. Books in SIT library.

```
mysql> SELECT b.Bid, b.Bname
-> FROM books b
-> JOIN ilibrary l ON b.Lid = l.Lid
-> WHERE l.lname = 'SIT Library';
```

Bid	Bname
201	Database Management Systems
202	Computer Networks
206	Introduction to Algorithms

3 rows in set (0.01 sec)

3. Books with less than 3 copies in SIT Library

```
mysql> SELECT b.Bname, n.no_of_copies
-> FROM books b
-> JOIN noofcopies n ON b.Bid = n.bid
-> JOIN ilibrary l ON n.blid = l.Lid
-> WHERE l.lname = 'SIT Library' AND n.no_of_copies < 3;
```

Bname	no_of_copies
Computer Networks	2

1 row in set (0.02 sec)

4. Sellers in the same city as SIT.

```
mysql> SELECT s.Sid, s.slname
-> FROM seller s
-> JOIN ilibrary l ON s.city = l.city
-> WHERE l.lname = 'SIT Library';
```

Sid	slname
501	Pragati Book Store
505	Pustak Seva
510	Mamta

3 rows in set (0.01 sec)

5. Sellers selling books to SIT.

```
mysql> SELECT DISTINCT s.slname
-> FROM seller s
-> JOIN purchase p ON s.Sid = p.Sid
-> JOIN ilibrary l ON p.Lid = l.Lid
-> WHERE l.lname = 'SIT Library';
```

slname
Pragati Book Store
Geeta
Pustak Seva

3 rows in set (0.01 sec)

6. Books authored by Brian Kernighan, and published by Tata McGraw Hill.

```
mysql> SELECT b.Bname
-> FROM books b
-> JOIN writes w ON b.Bid = w.Bid
-> JOIN author a ON w.Aid = a.Aid
-> JOIN publisher p ON w.Pid = p.Pid
-> WHERE a.Aname = 'Brian Kernighan' AND p.Pname = 'Tata McGraw Hill';
```

Bname
The C Programming Language

1 row in set (0.00 sec)

7. Books authored by Ken Thompson.

```
mysql> SELECT b.Bname
-> FROM books b
-> JOIN writes w ON b.Bid = w.Bid
-> JOIN author a ON w.Aid = a.Aid
-> WHERE a.Aname = 'Ken Thompson';
+-----+
| Bname                |
+-----+
| Operating System Concepts |
+-----+
1 row in set (0.00 sec)
```

8. Books issued by Mayank.

```
mysql> SELECT b.Bname
-> FROM books b
-> JOIN issue i ON b.Bid = i.Bid
-> JOIN student s ON i.memid = s.memid
-> WHERE s.Sname LIKE '%Mayank%';
+-----+
| Bname                |
+-----+
| Data Structures Using C |
+-----+
1 row in set (0.02 sec)
```

9. Books issued by SLS staff.

```
mysql> SELECT b.Bname
-> FROM books b
-> JOIN issue i ON b.Bid = i.Bid
-> JOIN staff st ON i.memid = st.memid
-> JOIN department d ON st.deptid = d.Deptid
-> WHERE d.Iname = 'SLS';
+-----+
| Bname                |
+-----+
| Cloud Computing Essentials |
+-----+
1 row in set (0.00 sec)
```

10. Publisher that provides books to SSBS through College Book Store.

```
mysql> SELECT DISTINCT pub.Pname
-> FROM publisher pub
-> JOIN purchase p ON pub.Pid = p.Pid
-> JOIN seller s ON p.Sid = s.Sid
-> JOIN ilibrary l ON p.Lid = l.Lid
-> WHERE l.lname = 'SSBS Library' AND s.slname = 'College Book Store';
+-----+
| Pname |
+-----+
| Wiley Publications |
+-----+
1 row in set (0.01 sec)
```

11. Institutes whose staff and students have issued book with bid 4444.

```
mysql> SELECT DISTINCT d.Iname
-> FROM department d
-> JOIN student s ON d.Deptid = s.deptid
-> JOIN issue i ON s.memid = i.memid
-> WHERE i.Bid = 4444
-> UNION
-> SELECT DISTINCT d.Iname
-> FROM department d
-> JOIN staff st ON d.Deptid = st.deptid
-> JOIN issue i ON st.memid = i.memid
-> WHERE i.Bid = 4444;
+-----+
| Iname |
+-----+
| SIT   |
+-----+
1 row in set (0.01 sec)
```

12. Sellers selling to libraries in the same city.

```
mysql> SELECT DISTINCT s.slname
-> FROM seller s
-> JOIN purchase p ON s.Sid = p.Sid
-> JOIN ilibrary l ON p.Lid = l.Lid
-> WHERE s.city = l.city;
+-----+
| slname |
+-----+
| Pragati Book Store |
| College Book Store |
| Vidyarthi Book Depot |
| Pustak Seva |
| Knowledge Hub |
+-----+
5 rows in set (0.00 sec)
```

13. Authors with books costing more than rs 500.

```
mysql> SELECT DISTINCT a.Aname
-> FROM author a
-> JOIN writes w ON a.Aid = w.Aid
-> JOIN books b ON w.Bid = b.Bid
-> WHERE b.Price > 500;
+-----+
| Aname |
+-----+
| Shruti |
| Ken Coel |
| Brian Kernighan |
| Shivam Kapoor |
| Ken Thompson |
| Thomas Cormen |
| Andrew Tanenbaum |
| Raghu Ramakrishnan |
| Elmasri Navathe |
+-----+
9 rows in set (0.00 sec)
```

14. Which members have issued books that belongs to E&TC department of SIT.

```
mysql> SELECT s.memid
-> FROM student s
-> JOIN department d ON s.deptid = d.Deptid
-> JOIN issue i ON s.memid = i.memid
-> WHERE d.deptname = 'E&TC' AND d.Iname = 'SIT'
-> UNION
-> SELECT st.memid
-> FROM staff st
-> JOIN department d ON st.deptid = d.Deptid
-> JOIN issue i ON st.memid = i.memid
-> WHERE d.deptname = 'E&TC' AND d.Iname = 'SIT';
+-----+
| memid |
+-----+
|   704 |
+-----+
1 row in set (0.00 sec)
```

15. List the books of SIT library

```
mysql> SELECT b.Bname
-> FROM books b
-> JOIN ilibrary l ON b.Lid = l.Lid
-> WHERE l.lname = 'SIT Library';
+-----+
| Bname |
+-----+
| Database Management Systems |
| Computer Networks |
| Introduction to Algorithms |
| The C Programming Language |
| Database Management Systems |
+-----+
5 rows in set (0.00 sec)
```

16. Which books are written by “Ken Coel”

```
mysql> SELECT b.Bname
-> FROM books b
-> JOIN writes w ON b.Bid = w.Bid
-> JOIN author a ON w.Aid = a.Aid
-> WHERE a.Aname = 'Ken Coel';
+-----+
| Bname |
+-----+
| Computer Networks |
+-----+
1 row in set (0.00 sec)
```

17. Find the staff members whose salaries are less than at least one employee.

```
mysql> SELECT DISTINCT s.stname, s.salary  
-> FROM staff s  
-> JOIN employee e ON s.salary < e.salary;
```

stname	salary
Satish Kumar	40000.00
Asha Menon	48000.00
Amit Joshi	45000.00
Priti Shah	58000.00
Rakesh Singh	52000.00
Anjali Rao	60000.00
Neha Sharma	50000.00
Sunil Patil	55000.00
Meera Nair	65000.00
Sanjay Gupta	70000.00

10 rows in set (0.03 sec)

18. Find the books whose price is greater than at least one book in the library.

```
mysql> SELECT DISTINCT b1.Bname, b1.Price  
-> FROM books b1  
-> JOIN books b2 ON b1.Price > b2.Price;
```

Bname	Price
Constitutional Law	13500.00
Business Analytics	1800.00
Cloud Computing Essentials	6500.00
Introduction to Algorithms	11000.00
Advanced Artificial Intelligence	12500.00
Data Structures Using C	1200.00
Software Engineering	799.00
Operating System Concepts	850.00
Database Management Systems	950.00
Database Management Systems	600.00
Computer Networks	750.00

11 rows in set (0.00 sec)

19. Find the employees who's salary is greater than all the employees of CS department.

```
mysql> SELECT e1.empname, e1.salary
-> FROM employee e1
-> WHERE e1.salary > ALL (
->     SELECT e2.salary
->     FROM employee e2
->     JOIN department d ON e2.Lid = d.Lid
->     WHERE d.deptname = 'CS'
-> );
```

empname	salary
Meena Joshi	50000.00
Rahul Shah	62000.00
Pooja Nair	48000.00
Sameer Kulkarni	70000.00
Farah Khan	52000.00
Deepak Verma	50000.00
Rekha Patil	47000.00
Ajay Mehta	81000.00

8 rows in set (0.01 sec)

20. Find such purchases who's total cost is greater than all the purchases which are done by
"SIT library"

```
mysql> SELECT p1.prid, p1.totalcost
-> FROM purchase p1
-> WHERE p1.totalcost > ALL (
->     SELECT p2.totalcost
->     FROM purchase p2
->     JOIN ilibrary l ON p2.Lid = l.Lid
->     WHERE l.lname = 'SIT Library'
-> );
```

prid	totalcost
1304	50000.00
1308	39000.00
1310	67500.00

3 rows in set (0.00 sec)

21. Find name of such SIU library which is located in Nashik and has Ilibrary named as “SIOM library”(solve using exists)

```
mysql> SELECT sl.lname
-> FROM siulibrary sl
-> WHERE sl.location = 'Nashik' AND EXISTS (
->     SELECT 1
->     FROM ilibrary il
->     WHERE il.Slid = sl.Slid AND il.lname = 'SIOM Library'
-> );
+-----+
| lname          |
+-----+
| SIU Nashik Libraries |
+-----+
1 row in set (0.03 sec)
```

22. Find such staff members who's name starts with “S” and do not work for “CS” department.
(solve using not exists)

```
mysql> SELECT s.stname
-> FROM staff s
-> WHERE s.stname LIKE 'S%' AND NOT EXISTS (
->     SELECT 1
->     FROM department d
->     WHERE s.deptid = d.Deptid AND d.deptname = 'CS'
-> );
+-----+
| stname          |
+-----+
| Sunil Patil     |
| Satish Kumar    |
| Sanjay Gupta    |
+-----+
3 rows in set (0.00 sec)
```

23. Which books are issued by student Shivani?

```
mysql> SELECT b.Bname
-> FROM books b
-> JOIN issue i ON b.Bid = i.Bid
-> JOIN student s ON i.memid = s.memid
-> WHERE s.Sname LIKE '%Shivani%';
+-----+
| Bname           |
+-----+
| Computer Networks |
+-----+
1 row in set (0.00 sec)
```

24. From which seller did SIT library purchases books of “Technical publications”?

```
mysql> SELECT DISTINCT s.slname
-> FROM seller s
-> JOIN purchase p ON s.Sid = p.Sid
-> JOIN ilibrary l ON p.Lid = l.Lid
-> JOIN publisher pub ON p.Pid = pub.Pid
-> WHERE l.lname = 'SIT Library' AND pub.Pname = 'Technical publications';
+-----+
| slname |
+-----+
| Pioneer Book House |
+-----+
1 row in set (0.00 sec)
```

25. Which books are issued by staff of SLS law department?

```
mysql> SELECT b.Bname
-> FROM books b
-> JOIN issue i ON b.Bid = i.Bid
-> JOIN staff st ON i.memid = st.memid
-> JOIN department d ON st.deptid = d.Deptid
-> WHERE d.Iname = 'SLS' AND d.deptname = 'Law';
+-----+
| Bname |
+-----+
| Cloud Computing Essentials |
| Constitutional Law of India |
+-----+
2 rows in set (0.00 sec)
```

26. Which sellers reside in the same city in which SCHC library resides?

```
mysql> SELECT s.slname
-> FROM seller s
-> JOIN ilibrary l ON s.city = l.city
-> WHERE l.lname = 'SCHC Library';
+-----+
| slname |
+-----+
| Knowledge Hub |
+-----+
1 row in set (0.00 sec)
```

27. Find out the publishers whose books are provided by the seller “pragati books store” to SSBS library.

```
mysql> SELECT DISTINCT pub.Pname
-> FROM publisher pub
-> JOIN purchase p ON pub.Pid = p.Pid
-> JOIN seller s ON p.Sid = s.Sid
-> JOIN ilibrary l ON p.Lid = l.Lid
-> WHERE s.slname = 'pragati books store' AND l.lname = 'SSBS Library';
+-----+
| Pname |
+-----+
| Oxford University Press |
+-----+
1 row in set (0.00 sec)
```

28. Which books are published by “Shivam Kapoor” with the publishing house Wiley publications?

```
mysql> SELECT b.Bname
-> FROM books b
-> JOIN writes w ON b.Bid = w.Bid
-> JOIN author a ON w.Aid = a.Aid
-> JOIN publisher p ON w.Pid = p.Pid
-> WHERE a.Aname = 'Shivam Kapoor' AND p.Pname = 'Wiley publications';
+-----+
| Bname |
+-----+
| Advanced Artificial Intelligence |
+-----+
1 row in set (0.00 sec)
```

29. What are the purchase details of the SIBM library for the month of December?

```
mysql> SELECT p.*
-> FROM purchase p
-> JOIN ilibrary l ON p.Lid = l.Lid
-> WHERE l.lname = 'SIBM Library' AND MONTH(p.purchase_date) = 12;
+-----+-----+-----+-----+-----+-----+-----+-----+
| prid | Lid | Sid | Pid | Bid | quantity | purchase_date | totalcost |
+-----+-----+-----+-----+-----+-----+-----+-----+
| 8884 | 102 | 884 | 884 | 884 | 20 | 2025-12-15 | 10000.00 |
+-----+-----+-----+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)
```

30. Which institute does the member belong who have issued book number 453?

```
mysql> SELECT d.Iname
-> FROM department d
-> JOIN student s ON d.Deptid = s.deptid
-> JOIN issue i ON s.memid = i.memid
-> WHERE i.Bid = 453
-> UNION
-> SELECT d.Iname
-> FROM department d
-> JOIN staff st ON d.Deptid = st.deptid
-> JOIN issue i ON st.memid = i.memid
-> WHERE i.Bid = 453;
+-----+
| Iname |
+-----+
| SIT   |
+-----+
1 row in set (0.00 sec)
```

Conclusion :

We successfully constructed, refined, and tested 30 complex SQL queries specifically tailored to your siu_library_db schema. Throughout this process, we effectively utilized various SQL techniques—including JOINS, UNIONS, subqueries, and EXISTS logic—to navigate a complex, multi-table library management system.